



Factor-based big data and predictive analytics capability assessment tool for the construction industry

Jasmine Ngo^a, Bon-Gang Hwang^{a,*}, Chenyue Zhang^{a,b}

^a Department of Building, National University of Singapore, 4 Architecture Drive, Singapore 117566, Singapore

^b TBH Global Services Pte Ltd, 8 Cross Street, #24-01, Manulife Tower, Singapore 048424, Singapore

ARTICLE INFO

Keywords:

Big data
Predictive analytics
Capability assessment tool
Construction industry
Organization capability

ABSTRACT

Big data and predictive analytics have huge potential to create value to the construction industry. However, there is a lack of benchmarking system to evaluate organizations' competency to adopt big data and predictive analytics. Hence, this study aims to develop a big data and predictive analytics capability assessment tool that can measure construction organizations' capability in big data and predictive analytics implementation and that also highlights strengths and weaknesses of the organization to provide a benchmark in the process of big data and predictive analytics implementation. 21 determinants were identified and assessed in sense of their impacts on an organization's capability to implement big data and predictive analytics. These determinants were categorized into five determinant groups and assigned weights, to form the basis for the big data and predictive analytics capability assessment tool. The developed tool was then validated with four construction organizations to reflect their big data and predictive analytic capability levels, strengths and weaknesses. The findings of this study contribute to knowledge and practice by identifying the determinants impacting construction organizations' capability to adopt big data and predictive analytics and in the development of a computerized assessment tool which also serves as a benchmarking tool for construction organizations in the implementation of big data and predictive analytics.

1. Introduction

Big Data Analytics (BDA) refers to the application of a series of analytical techniques on large data sets to identify patterns and derive insights for business performance. Since late 2011, big data (BD) has become increasingly popular as evident in its high popularity on Google Trends with average interest index above 80 out of 100. The advancement of the Information Technology (IT) industry, the growing number of the Internet-of-Things (IoT), the proliferation of affordable data storage and processing technologies lead to the rise of BD. The BD market is forecasted to be more than doubled of its expected market size in 2018 and its market value is also expected to reach US\$103 billion in 2027 [1].

While Big Data Engineering (BDE) provides supporting functions like data storage and processing for analytics [2], BDA is mainly related to pattern identification and value creation [3]. Common types of analytics include descriptive analytics, diagnostic analytics, predictive analytics and prescriptive analytics in the order of ascending value and complexity.

With increasing amount of data generated at high velocity from

various sources in the construction industry and the advancements in data analytics techniques with capabilities to deal with extremely large data sets, there is great potential for the construction industry to benefit from big data and predictive analytics (BDPA) throughout the project lifecycle. BDPA refers to the application of predictive analytics on BD. BDPA can be used as a decision support tool, such as in the selection of optimal site location [4], in predicting project delays [5], and to predict energy consumption of buildings [6].

Despite the variety of BDPA applications in the construction industry and great potential to create value for the construction industry throughout the project lifecycle, the adoption of BDPA in the construction industry remains at a nascent stage [7]. In addition, the construction industry is known as a laggard in technology adoption and have a strong general resistance towards new technologies [8]. The integration of BDPA techniques, technologies and processes into the existing organization culture, systems and processes is critical to maximize the potential of BDPA [8]. Hence, for the potential of BDPA to be maximized, there is a need to evaluate the competencies and readiness of organizations to adopt BDPA.

However, there are limited BDPA studies that focus on the

* Corresponding author.

E-mail addresses: jasminengo@u.nus.edu (J. Ngo), bdghbg@nus.edu.sg (B.-G. Hwang).

<https://doi.org/10.1016/j.autcon.2019.103042>

Received 6 June 2019; Received in revised form 5 October 2019; Accepted 30 November 2019

Available online 09 December 2019

0926-5805/ © 2019 Elsevier B.V. All rights reserved.

construction industry. At the same time, there is a lack of benchmarking system to evaluate organizations' competencies and readiness to adopt BDPA. As a result, this study aims to answer the following research questions – (i) What are the key determinants that impact construction organizations' capability to implement BDPA; (ii) Do construction organizations have the capability and readiness to implement BDPA currently; and (iii) How can an automated assessment tool be developed for organizations to assess their capability to implement BDPA. To develop a BDPA Capability Assessment Tool (CAT) that can measure construction organizations' capability in BDPA implementation and that also highlights strengths and weaknesses of the organization provide a benchmark in the process of BDPA implementation, the key determinants that impact construction organizations' capability to adopt BDPA will first be explored, followed by a general assessment on construction organizations' existing capability and readiness to implement BDPA. Finally, the determinants will be assigned weights using the analytical hierarchical process (AHP), which will form the basis for the BDPA-CAT.

The findings of this paper contribute to the body of knowledge and practice by identifying the determinants that impact construction organizations' capability to adopt BDPA, assessing the existing capabilities of construction organizations to adopt BDPA and in the development of BDPA-CAT, an automated assessment tool to assist in organizational processes involved in the initiating, planning, execution, monitoring and control of implementing BDPA. The BDPA-CAT has several purposes namely – (i) assessment of construction organization's organizational readiness to adopt BDPA; (ii) assessment of construction organization's capability to adopt BDPA to identify areas of improvement and strengths; and (iii) benchmarking tool for construction organizations and their competitors. The BDPA-CAT serves as a quick, consistent and accurate assessment tool for construction organizations to assess their BDPA capability. The findings further complement existing studies that focuses on advancing the technical processes of BDPA in the construction industry by providing a holistic benchmarking system for organizations looking to implement BDPA.

2. Background

2.1. Big Data and predictive analytics

BD is commonly associated with the 3Vs, namely – volume, velocity and variety, where the size and complexity of the datasets have exceeded the capability of conventional database software tools to capture, store, manage and analyze [9]. Volume refers to the magnitude of the data; velocity refers to the data generation rate and data delivery rate; and variety refers to the structural heterogeneity in a dataset [10,11]. Hence, new methods and innovative solutions are required to store, process and analyze the large and complex datasets.

There are two main parts in BD, namely – BDE and BDA. BDE is concerned with the technologies required to store, manage and process data. Examples of BD storage include distributed file system and NoSQL databases and BD processing include MapReduce and Spark [7]. On the other hand, BDA is the process of applying analytics on large datasets to extract knowledge to support decision making [12]. There is a growing trend for organizations to collect data from various resources, analyze them to form strategies and to improve business performance. Its wide applications cover multiple perspectives of the construction industry, including supply chain, design, planning, monitoring, control, and energy management.

Data analytics is typically categorized as – (i) descriptive analytics; (ii) diagnostic analytics; (iii) predictive analytics; and (iv) prescriptive analytics. The value contribution and complexity of each type of analytics is presented in ascending order. The four categories of data analytics apply to both traditional data analytics and BDA. The key distinction between traditional data analytics and BDA mainly lies in the volume and types of data that are analyzed, where BDA can be

applied to structured, semi-structured and unstructured data as compared to structured data for traditional data analytics [13,14]. Structured data refers to data that is clearly defined, organizations in a standardized way and usually stored in relational databases while unstructured data refers to data that contain information but does not have an explicit structure [15,16]. Common unstructured data include images, emails, websites and reports [15,16].

Descriptive analytics are used to categorize, characterize, consolidate and classify data to generate valuable information for the purpose of identifying and understanding problems in organizations' performance. Descriptive analytics are the most frequently used and easily perceived type of analytics [17]. Diagnostic analytics evaluates the potential causes of a problem. Predictive analytics makes use of statistical and data mining techniques to predict the future by analyzing past performance through examining historical data, identifying patterns and extrapolating these relationships. Prescriptive analytics applies algorithms on datasets to explore and examine potential decisions that involve high-volume and complex objectives and requirements in order to optimize organizational performance [18].

Currently, most organizations make use of descriptive analytics for better understanding of what happened. Some organizations use diagnostic analysis to identify the root causes and isolate confounding information and a few data-driven organizations go as far as predictive analytics to make insightful forecasts. Furthermore, due to the growing volume, velocity and variety of data, more affordable and powerful computers, user-friendly software and strategic needs for differentiation, the use of predictive analytics has been increasing. On the other hand, there are few notable examples of good “prescriptive” analytics in the real world due to insufficient number of dimensions that databases capture. The analysis from such data is unable to provide mature and feasible insights for complex business problems [19]. Thus, this paper places emphasis on predictive analytics that can be applied on BD as it is attainable with the current state of data analytics techniques while adding more value than descriptive and diagnostic analytics.

2.2. BDPA techniques

Statistical models form the foundation to predictive analytics for application with normal data sets. However, when predictive analytics is applied to large data sets, a few issues are worth attention compared to the application conventional statistical methods on small data sets. Firstly, the concept of statistical significance is less relevant to big data because big data samples are massive and may be able to represent the majority or the entire population. Secondly, due to the complex nature of BD, conventional statistical methods may not be easily and efficiently applied. Hence, new statistical methods for BDPA are in demand [20]. However, BDPA techniques are relatively new and are still being developed [7]. These BDPA techniques are built on existing predictive analytics method. Hence, the predictive analytics techniques introduced are existing predictive analytics methods, which can further be extended to analyze BD.

Common predictive analytics techniques belong to the broader set of machine learning techniques. Machine learning can be categorized into supervised learning and unsupervised learning. Unsupervised learning aims to describe association and patterns among a set of input measures with no defined outcome. Typical examples include clustering, natural language processing and information retrieval. These are typically used for descriptive and diagnostic analytics to inform users about what happened and to identify the root causes of problems. Supervised learning, which includes regression and classification, aims to predict the outcome based on a number of input values [21]. This is similar to predictive analytics, which is used to predict the future (“Outcome”) based on the past data (“input”) and hence belong to supervised learning. Therefore, this paper focuses on regression and classification techniques.

Regression is a technique to predict the target variable's numerical

Table 1
Determinants impacting construction organizations' BDPA capability.

Code	Determinants	Relevant literature
DG1	Determinant Group 1 - Data Characteristics	
D1	Data volume – The amount of data is large enough for BDPA.	[10]
D2	Data velocity – Frequency of organization's incoming data that requires processing is high.	[10]
D3	Data variety – Organization's data comes from a variety of sources.	[10]
D4	Data veracity – Organization's data is reliable.	[14,38]
DG2	Determinant group 2 - Organization operations	
O1	Clear Standard Operating Procedure (SOP) – There is a clear plan to implement BDPA.	[39]
O2	Financial Capability – The budget is sufficient to implement BDPA	[40–42]
O3	Sufficient Time – There is enough time for BDPA development.	[40–42]
O4	Dedicated Team – There is a dedicated team/function in charge of BDPA.	[39]
O5	Regular Review – There are regular review sessions to evaluate BDPA performance.	[39]
DG3	Determinant Group 3 - Technology and Software	
T1	Data Visualization Tool – There is a visualization tool such as a dashboard for users to visualize the analysis processes.	[40,43]
T2	Cloud-based Platforms for Data Processing and Analytics	[40,43]
T3	Open Source Software for BDPA	[40,43]
T4	New Forms of Databases for Data Storage	[40,43]
DG4	Determinant Group 4 - Technical Skills and Expertise	
S1	Technical Support – Users can rely on technical support while facing technical issues.	[39]
S2	Training – The organization provides necessary BDPA training (either internal or external) to its employees.	[40,42,44]
S3	Skill Sets and Competency – The overall skill sets and competency of BDPA staff in the organization is competitive.	[40,42,44]
DG5	Determinant Group 5 - Organization Culture	
C1	Data-driven culture	[45,46]
C2	Top management support for innovation	[47]
C3	Alignment with organization's vision	[47]
C4	Continuous improvement	[40,48]
C5	Adaptability to change	[49]

value based on input variables. Common variations of regression include simple linear regression which models the relationship between a dependent variable and an independent variable, multiple linear regression models which models the relationship of one dependent variable and two or more independent variables, and logistic regression which models the relationship between categorical dependent variables and one or more independent variables.

Classification is the process of emulating decisions automatically through trained data sets [7]. Prominent classification algorithms include decision tree which utilizes a tree structure to demonstrate the process of deriving the decisions and final outcomes [22], Naïve Bayes which computes the joint probabilities of values with categorical attributes [22], neural networks which handles more complex non-linear relationships in large sets of training data [23], K-nearest neighbor which predicts an object's value or class memberships based on the k-closest training samples, support vector machine which combines linear models with instance-based learning techniques to construct a linear decision function [22], random forest which combines several tree predictors to overcome the issue of overfitting using decision trees, time series forecasting for data collected over time, and partial least squares to development predictive models in the presence of multiple and highly collinear factors.

2.3. BDPA applications in the construction industry

Although BDPA applications are not yet widely adopted in the construction industry [7], demonstrate great potential to improve the performance of the construction industry throughout the project lifecycle. Furthermore, a variety of data is available in the construction industry including project schedules, project drawings, project site images, project costs, progress reports and sensor data [7,24]. In addition, with the advancement in Building Information Modeling (BIM), sensor technologies and Internet-of-Things, the volume, velocity and variety of data is expected to grow [7]. Valuable knowledge can be extracted from these data to support decision-making and improve the performance of the construction industry throughout the project lifecycle, which further increases the benefits of utilizing BDPA in the construction industry [12]. The following are the various applications

according to the stages of a project lifecycle.

In the pre-tender stage, BDPA can be used to support decision-making in site selection, synthesizing demographic, economic and labour information to generate intelligent location selection decisions for developers [4]. In the tender evaluation stage, BDPA can be used in tender price evaluation [25] and failure prediction model for construction firms [26]. In the construction stage, BDPA can be used for prediction of early cost estimation accuracy for construction projects [27], prediction of cost overrun conditions [28], project delay prediction [5], minimize unsafe worker behavior [29], construction waste management in terms of waste prediction, visualization and minimization [30], and construction site management through building information modeling (BIM) information analytics [31–33]. In the operations and maintenance stage, BDPA can be used in energy management to assess if the building has complied with energy saving targets, to predict the energy consumption of a building based on historical trends, and to predict building use type, performance class and operational behavior [6,34–37].

2.4. Determinants impacting construction Organizations' capability to adopt BDPA

To maximize the potential of BDPA, in addition to the availability of BD, organizations' processes, systems and culture must be able to support the integration of BDPA technologies, techniques and processes [7,8]. Hence, a total of 21 determinants that impact organizations' capability to adopt BDPA have been identified and categorized into five determinant groups (DG): DG1 - Data Characteristics; DG2 - Organization Operations; DG3 - Technology and Software; DG4 - Technical Support and Expertise; DG 5 - Organization Culture as shown in Table 1.

The success of BDPA is tightly linked to the data characteristics (DG1) of an organization. The 3Vs Framework proposed by Laney (2001) summarized the most important traits of big data. It is essential for an organization to have large volume (D1) of unstructured data from various sources (D3) at high velocity (D2) so that patterns can be recognized for predictive analytics. In addition, data integrity (D4) is crucial to ensure the reliability of the input data as it would impact the

value of the analysis results.

A meticulously planned organization operations (DG2) routine is also important to maximize the organization's competency. A clear standard operating procedure (SOP) should be defined and communicated to all the stakeholders within the organization (O1). Besides that, tangible resources like budget (O2) and time (O3) are indispensable for the development of BDPA. A dedicated team, department or sub-function should be formed to take charge of BDPA related activities (O4). Overall, regular reviews of the BDPA operations are necessary to streamline and processes and raise effectiveness (O5).

Moreover, technological development is always related to technology, software (DG3), technical skills and expertise (DG4). DG3 specifically looks into software related issues. It includes the presence of a visualization tool for better user interface (T1), cloud-based platforms for data processing and analytics (T2), open source software for data analytics (T3) and new generations of database for massive data storage (T4). Comparatively, DG4 is more related to BDPA talent development and training. It is concerned with the availability of BDPA talents (S3), trainings provided by organizations (S2) and BDPA related technical support during daily operations (S1).

Furthermore, organization culture (DG5) sets the long-term directions for any organizations. The most relevant cultural determinant for BDPA development is whether an organization has a data-driven culture (C1). The decision-making process is tightly linked to the data analysis results [43]. Top management should take the initiative to explain the benefits and potential competency that BDPA may bring to the organization. Top management should also be able to form high-level strategies for BDPA development and monitor its performance (C2). Moreover, the level of acceptance and adoption of BDPA is also related to the organizations' vision as it will determine the priority of BDPA related activities (C3). As BDPA technologies evolve at a high speed, organizations with a "continuous improvement/continuous learning" attitude would build their competitiveness and benefit more from BDPA (C4). Lastly, adaptability to change in this ever-changing world enables organizations to keep aware of the latest market trends and government policies, regulations and incentives so that their practices will not be obsolete (C5).

3. Research method and data presentation

The research process consists of six steps. In Step 1, a comprehensive literature review was conducted to establish a foundation for this study and to support the development of a survey questionnaire. The comprehensive literature review conducted identified 21 determinants that impact construction organizations' capability to implement BDPA as presented in Table 1. Based on the findings from Step 1, a survey questionnaire was developed and validated through pilot interviews which were carried out with three industry experts with more than 10 years of relevant experience in Step 2. The pilot interviews were conducted to validate the set of determinants that impact construction organizations' capability to implement BDPA and the validity and relevance of the survey questionnaire. Pilot interviews have been used in research studies in the construction industry as a validation tool of survey questionnaires [50,51]. Step 3 was to administer the survey questionnaire to industry practitioners to assess if construction organizations have the capability to implement BDPA based on the determinants identified and to serve as a reference for the perceived importance of determinant groups during the development of the scoring approach.

The finalized questionnaire consisted of six sections. Part I asks about the respondent's background including name (optional), position, years of experience in the construction industry and years of applying predictive analytics respectively. Part II asks about the background information of respondent's organization - name of organization, business domain, years of experience in the construction industry and years of experience of practicing predictive analytics. Part III focuses on the

data characteristics, technological needs to store and process data and analytical methods. Part IV asks about organizational resources, procedures and strategies related to BDPA implementation. Part V requires respondents to rank the importance of determinant groups. Part VI consists of an open-ended question to ask respondents about their suggestions and comments to this study and invites them to participate the follow-up study. The capability of construction organizations to implement BDPA according to the determinants are assessed using a five-point Likert scale for most determinants, excluding D1, T1, T2, T3, T4 and S3.

The target respondents were relevant professionals working in Singapore's construction industry, including project managers, facility managers, quantity surveyors, engineers, architects, designers, government officers and data analytics related role in construction firms. The organization's business domains include consulting firms, developers, contractors, architectural design, facility or property management and government authorities.

Based on literature review and exploratory studies, only firms up to a certain scale may accumulate large amount of data and possess resources to implement BDPA. Therefore, non-probabilistic sampling was used to determine the sample frame. The scale of construction firms, the experience and positions of survey respondents were reviewed during sample selection. The following websites were used as primary sources to search for survey respondents: BCA Directory of Registered Contractors and Licensed Builders, the Singapore Institute of Surveyors and Valuers, Singapore Directory of Professional Engineers, International Facility Management Association Singapore and the Singapore Government Directory.

The main survey was conducted through an online survey platform. One of the benefits of online surveys is that respondents can complete the survey at their own paces and through various mobile devices. Unlike the more socially desirable responses collected in a telephone survey, more reliable responses can be collected through online survey. Web links and QR code to the survey were embedded in the emails which were disseminated to 170 survey targets in January and early February 2019. A total of 42 responses were received and the survey response rate is 24.7%. The response rate is within the acceptable range for survey practitioners [52]. Furthermore, as the sample size of this study of 42 is greater than 30, statistical analysis could be performed because the central limit theorem holds true for sample sizes larger than 30 [53,54]. The profiles of the respondents and their organizations are presented in Table 2.

The profile of organizations covers major business domains in the construction industry such as government agencies, developers, contractors, consultants and property management firms. In addition, the respondents were from different occupational backgrounds including project manager, architect, engineer, quantity surveyor, facility manager and government officer. The diversity of the respondent panel provides reliable and meaningful survey result analysis.

The survey results were analyzed through Statistical Package for the Social Sciences (SPSS). To determine if parametric or non-parametric methods should be adopted for data analysis, the Shapiro-Wilk test was used to test the normality of the sample [55]. Using a chosen alpha level of 0.05, the *p*-value generated from the test indicated that the data tested are not normally distributed, and hence, non-parametric methods would be adopted for the subsequent data analysis [55]. The organizations' capability to implement BDPA was assessed using mean rank analysis. The responses from the respondents for the determinants were used to calculate the mean score for the determinants. Mean scores above the mean of 2.5 were considered to be representative of the organizations having moderate capability in the determinant. The ranking of the determinant groups were based on the mean scores [56]. To assess the significance of the determinant groups the one-sample Wilcoxon signed-rank test was adopted. The one-sample Wilcoxon signed-rank test is the non-parametric method of conducting the one-sample *t*-test. The one-sample *t*-test is used to assess if the identified factor is

Table 2
Profile of the respondents and organizations.

		Frequency	Percentage (%)
Respondent's position	Project manager	14	33.3
	Facility manager	8	19.0
	Engineer	7	16.7
	Quantity surveyor	5	11.9
	Government officer	4	9.5
	Architect	3	7.1
	Others	1	2.4
Years of experience in the construction industry	0–10	15	35.7
	> 10	27	64.3
Years of experience using predictive analytics	0–5	23	54.8
	> 5–10	10	23.8
	> 10	9	21.4
Total		42	100.0
Organization' business domains	Contractor	12	28.6
	Consultant	8	19.0
	Property management firm	8	19.0
	Quantity surveyor	5	11.9
	Developer	4	9.5
	Government agency	4	9.5
	Others	1	2.4
	1–20	18	42.9
	> 20	24	57.1
	0–10	29	69.0
Years of experience using predictive analytics	> 10	13	31.0
	Total	42	100.0

statistically significant [56]. The Kruskal-Wallis one-way analysis was used to compare if there are statistical differences between two or more groups [57].

Using the data collected by the survey questionnaire as a reference, a BDPA capability scoring approach was developed in Step 4 to assess the capability of construction organizations in BDPA implementation. The development of the BDPA capability scoring approach is detailed in Section 5.1. Based on the BDPA capability scoring approach, the BDPA-CAT was developed using Microsoft Excel and Microsoft Visual Basic for Applications (VBA) in Step 5. Microsoft Excel was selected as the platform for the BDPA-CAT as the software is widely used in the industry and accessible to the public [58]. VBA has been optimized to support the development of applications [59]. The BDPA-CAT will also be updated automatically whenever there is an update on Microsoft Excel, which reduces the need for separate maintenance and upgrading of the BDPA-CAT [59].

Lastly, the BDPA-CAT was validated in Step 6 with industry practitioners. The practitioners were identified in the process of identifying the target respondents for the survey questionnaire. Four professionals with at least ten years of experiences in the construction industry were interviewed to validate the BDPA-CAT. The organizations had rich experiences in the construction industry. The four professionals and organizations are independent from the respondents in the survey and were asked to rate their organizations on the 21 determinants. Since the respondents have not been exposed to the determinants beforehand, fresh insights and more reliable responses can be collected. Table 3

Table 3
Profile of organizations and interviewees for BDPA-CAT validation.

Organization	Business Domain	Years of experience in the construction industry	Organization adopts predictive data analytics	Interviewee	Position	Years of Experience in the construction industry	Uses predictive data analytics
A	Consultant	20	Yes	D	Project manager	12	Yes
B	Contractor	18	No	E	Engineer	15	Yes
C	Consultant	> 20	Yes	F	Cost manager	10	Yes
D	Developer	> 20	Yes	G	Project manager	16	Yes

Table 4
Results for mean rank analysis and Shapiro-Wilk Test for various determinants.

Determinant group	Code	Mean	Shapiro-Wilk test
DG1 – Data characteristics	D2	3.738	0.000
	D3	3.857	0.000
	D4	3.357	0.000
DG2 – Organization operations	O1	3.167	0.001
	O2	3.048	0.001
	O3	3.143	0.002
	O4	2.524	0.001
	O4 _{se}	3.111	0.032
DG4 – Technical skills and expertise	O5	3.238	0.002
	S1	3.310	0.002
	S2	3.476	0.001
DG5 – Organization culture	C1	4.024	0.000
	C2	4.190	0.000
	C3	3.786	0.000
	C4	4.381	0.000
	C5	4.286	0.000

* Organizations with no existing data analytics team were directed to indicate if the organization has plans to form such a team in the near future.

show the profile of organizations and interviewees for the BDPA-CAT validation. Interviewees D, E, F, G are from companies A, B, C, D respectively and provided first-hand information about their organizations during the BDPA-CAT validation stage. The results of the analysis indicated their organizations' performance in building up the capability to implement BDPA.

4. Data analysis and discussion

The organizations' capability to implement BDPA according to the 21 determinants were assessed based on the responses of the survey questionnaire. As DG3, D1 and S3 were more appropriately assessed without the use of Likert Scale, the results of these determinants were not included in Table 4. The summary of the mean rank analysis and Shapiro-Wilk tests for the remaining determinants are presented in Table 4.

4.1. Determinant group 1 - data characteristics

As shown in Fig. 1, majority of construction firms recognize that their volumes of data suitable for analytics are in the range of 10–100 TB (42.9%) or 1–10 TB (26.2%) at present. Smaller data volume below 1 Tb (4.8%) or extremely large data volume above 1Pb (2.4%) are rarely observed. Comparing Today's figures with the five-year forecasts, there is a noticeable trend that the volume of data for analytics shifts towards larger data range such as 100–500 TB (increase from 11.9% to 16.7%). With the advancement in technology such as IoT, sensors and digitalization transformation goals, the growth in data volume for analytics for construction firms is optimistically foreseeable.

The respondents largely indicate that there is sufficient data velocity, variety and veracity in their organizations with the means 3.738, 3.857 and 3.357 respectively. In addition, the respondents were asked to indicate the percentage of unstructured data in their organizations.

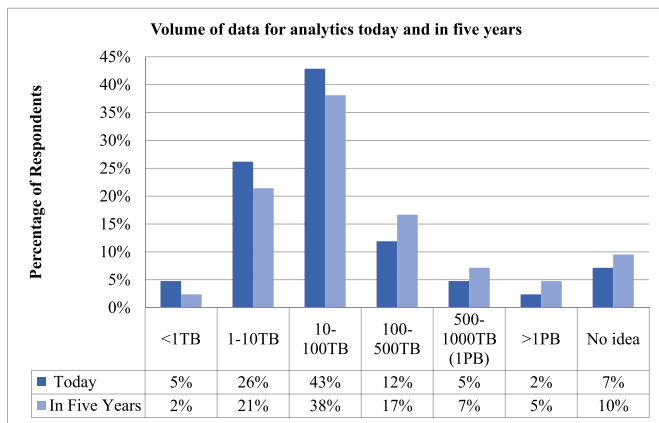


Fig. 1. Volume of data for analytics today and in five years.

Percentage of Unstructured Data

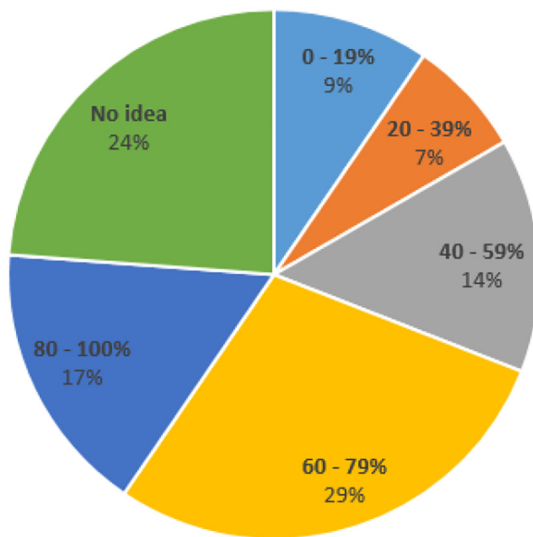


Fig. 2. Percentage of unstructured data.

As seen in Fig. 2, about half (45.3%) of the respondents indicated that their organizations have more than 60% of unstructured data. As BDPA allows organizations to extract valuable information from unstructured data, the availability of unstructured data demonstrates the potential of BDPA in generating value for the construction industry [13,14].

4.2. Determinant group 2 - organization operations

The results indicate that organizations' operations are moderately suitable for BDPA implementation with the means 3.167, 3.048, 3.143, 2.524, and 3.238 for determinants O1 to O5 respectively. It was found that the overall mean of DG2 is relatively lower compared to the other DGs. This is reasonable as existing organization operational processes are established for the organization's current data management processes, which may or may not involve BD. Furthermore, the adoption of BD is still in its nascent stage, as evident in the responses of O4, with a mean of 2.524 [7].

4.3. Determinant group 3 - technology and software

Under the Technology and Software section, respondents were asked whether their organizations had used any dedicated software or

platforms for data storage and processing and were required to indicate the technology and software used by their organizations by selecting from a list of software provided in the questionnaire. The list of data processing software included Apache Spark, Amazon EMR, Cascading, Hive, MapReduce, Microsoft Excel, Polybase, Presto, R, Scribe, Sqoop and an option for others. The list of data storage software included Apache Cassandra, Apache CouchDB, Apache HBase, Hadoop Distributed File System, Microsoft HDInsight and an option for others.

Survey results indicated that 64.3% of the construction firms do not adopt new forms of databases or data warehousing technologies. Project data are stored separately and searching historic data for decision making is time-consuming. The remaining 35.7% of the respondents indicate that their organizations put in efforts to data storage management.

In terms of data processing, Microsoft Excel (88.1%) is the top ranked technology from all survey respondents. Besides the list of technologies provided in the survey questionnaire, some respondents supplemented other non-proprietary software adopted by their organizations, such as IBM SPSS Predictive Analytics tools, SAP Data Services and SAS Analytics Suite. 38.1% of the firms showed attempts to use cloud-based platforms for data processing and analytics.

4.4. Determinant group 4 - Technical skills and expertise

The results indicate that organizations have moderate availability of technical support and provision of structured data analytics training with the means 3.310 and 3.476 respectively.

In terms of skill sets and competency, a mean rank analysis was performed to analyze the top three most popular analytical techniques by the years of experience of using predictive data analytics, from 0 to 5 years, > 5 years to 10 years, and > 10 years' experience for respondents, and 0 to 10 years, and > 10 years for organizations, shown in Table 5.

The most popular analytical technique was found to be simple linear regression with 90.48% of the respondents and 42.86% of the organizations that use the technique frequently or very frequently. Decision tree is also a popular choice that appears among the top three choices across the groups with 71.43% of the respondents and 42.86% of the organizations that use the technique frequently or very frequently. Multiple linear regression is another technique which 80.95% of the respondents and 42.86% of the organizations that have used the technique, but less frequently than simple linear regression and decision trees. Respondents and organizations with more experience favour time series forecast. Organizations prefer common analytical methods such as regression instead of complex machine learning techniques. This may be due to the lack of tangible resources for research and implementation, reluctance by management to invest in innovation, and lack of perceived return on investment [8].

4.5. Determinant group 5 - Organization culture

The existing organization culture appears to be suitable for BDPA implementation with the means 4.024, 4.190, 3.786, 4.381 and 4.286 for C1 to C5 respectively.

4.6. Ranking of determinant groups

Wilcoxon Signed Rank Test was performed to assess the importance of the determinant groups. Due to the 5-point Likert Scale, the mean for the null hypothesis was set at 3. All the determinant groups are considered important to contribute to an organization's BDPA capability. Subsequently, the Kruskal-Wallis Test results indicate that respondents with various positions reach a consensus on the importance of the determinant groups. The results are shown in Table 6.

Through the mean rank analysis, the order of importance from the most important to the least important for the five determinant groups is

Table 5
Mean rank analysis of predictive analytics methods adopted.

Analytical methods	Years of experience of using predictive data analytics (Respondents)						Years of experience of using predictive data analytics (Organizations)			
	0–5 years		5–10 years		> 10 years		0–10 years		> 10 years	
	Mean	Rank	Mean	Rank	Mean	Rank	Mean	Rank	Mean	Rank
Simple linear regression	3.92	1	4.25	1	4.31	1	3.98	1	4.04	1
Multiple linear regression	3.85	2	3.79	4	4.03	4	3.65	3	3.3	4
Logistic regression	3.14	6	3.02	7	3.23	7	2.31	5	1.13	7
Decision tree	3.65	3	4.05	2	3.93	3	3.76	2	3.89	2
Naïve Bayes	2.33	8	3.75	5	3.87	5	1.22	7	1.31	5
Artificial Neural Network (ANN)	2.03	10	2.56	11	2.88	9	1.00	9	1.05	8
K-nearest Neighbor (KNN)	2.43	7	2.78	9	2.74	10	1.00	9	1.00	10
Support Vector Machine (SVM)	2.27	9	2.58	10	2.43	11	1.00	9	1.00	10
Random forest	3.19	5	3.12	6	3.14	8	1.05	8	1.05	8
Time series forecast	3.45	4	3.87	3	4.06	2	3.13	4	3.54	3
Partial Least Squares (PLS)	1.98	11	2.96	8	3.59	6	2.19	6	1.27	6

Rating scale: 1 - do not adopt; 2 - use the technique rarely; 3 - use the technique occasionally; 4 - use the technique frequently; 5 - use the technique very frequently.

as follows: Organizational Culture (DG5), BDPA Operations (DG2), Data Characteristics (DG1), Technical Skills and Expertise (DG4) and Technology and Software (DG3). This result indicates that non-technical determinants are weighed more than technical determinants. This is reasonable as technical aspects can be outsourced or acquired externally while intangible aspects like organization processes and culture are harder to be transformed. Furthermore, this supports the view of previous researchers that stated that technologies can only be as effective as its integration in an organization's processes and culture [8,60].

5. Development of BDPA-CAT

The survey results indicate that construction organizations have the capability to implement BDPA. However, the level of adoption for BDPA remains low. Hence, the BDPA-CAT was developed with the aim to increase adoption of BDPA in the construction industry by providing a benchmark for construction organizations looking to implement or has implemented BDPA, in the assessment of its capabilities and to highlight its strengths and areas for improvement.

5.1. BDPA capability scoring approach

The five determinant groups were found to be important in its contribution to an organizations' BDPA capability as found in the One-sample Wilcoxon Signed Rank Test and no statistical differences were found between respondents of different positions. However, the magnitude of contribution of each determinant group and each determinant to the organizations' BDPA capability remains unknown. In order to derive the weightage, analytical hierarchy process (AHP) was adopted. AHP can be applied to transform various types of criteria into numerical values to facilitate decision making and achieving optimal result [61].

The first stage of AHP involves pairwise comparisons to assign weightages for the five determinant groups. Since there are five determinant groups, a matrix of 5×5 is set up for the pairwise

comparison between any two determinant groups. A value between one and five is assigned to the more important DG of the pair. The other less important DG will be assigned the reciprocal of that value. The denotation of the scale from 1 to 5 is shown below:

- 1: The two determinants contribute equally.
- 2: One determinant is slightly favoured over the other.
- 3: One determinant is moderately favoured over the other.
- 4: One determinant is strongly favoured over the other.
- 5: One determinant dominates over the other.

After the completion of all the pairwise comparisons, sum of each column will be computed at the last row of the table. The values are then normalized by dividing the value in each cell over the column sum. In order to obtain normalized weightages, the value in each cell is divided by the column sum. Row sums and average of each row can then be computed. AHP is performed on determinants under each DG in the same routine. Finally, numerical weightages will be assigned to all the determinants, as shown in Table 7.

The computation of the BDPA-CAT score can be illustrated by the following formula:

$$S_{BDPA-CAT} = 20 \cdot \sum F_{DG} (K_d \cdot S_d) \quad (1)$$

The user input for each determinant will be on a scale of one to five while the output of the BDPA-CAT score will be on a scale of one to one hundred for easy interpretation. Hence, the fixed coefficient 20 will be multiplied. K_{DG} refers to the weightage of each determinant group while K_D refers to the weightage of each determinant within the group. S_D refers to the user input score for each determinant on a scale of one to five and $S_{BDPA-CAT}$ refers to the output score on a scale of one to one hundred. With the finalized weightages and user input score for each determinant, a total score can be computed by applying Eq. (1).

Table 6
Importance and ranking of determinant groups.

Determinant groups	Mean	Rank	Shapiro-Wilk test	One-sample Wilcoxon signed rank test	Kruskal-Wallis test (by respondents' positions)
DG1	3.91	3	0.000	0.000	0.992
DG2	4.12	2	0.000	0.000	0.751
DG3	3.62	5	0.001	0.001	0.990
DG4	3.79	4	0.000	0.000	0.833
DG5	4.26	1	0.000	0.000	0.911

Table 7
Final normalized weightages of determinants and determinant groups.

Determinant groups	Code	Determinants	Finalized weightage
DG1 - Data characteristics (0.189)	D1	Data volume	0.338
	D2	Data velocity	0.219
	D3	Data variety	0.269
	D4	Data veracity	0.174
DG2 - Organization operations (0.242)	O1	Clear SOP	0.155
	O2	Financial capability	0.311
	O3	Sufficient time	0.311
	O4	Dedicated team	0.124
DG3 - Technology and software (0.117)	O5	Regular review	0.098
	T1	Data visualization tool	0.167
	T2	Cloud-based platform for data processing and analytics	0.313
	T3	Open source software for BDPA	0.238
DG4 - Technical skills and expertise (0.152)	T4	New forms of databases for data storage	0.282
	S1	Technical support	0.197
	S2	Training	0.279
	S3	Skill sets and competency	0.524
DG5 - Organization culture (0.300)	C1	Data-driven culture	0.240
	C2	Top management support for innovation	0.240
	C3	Alignment with organization's vision	0.140
	C4	Continuous improvement	0.180
	C5	Adaptability to change	0.201

5.2. BDPA-CAT architecture

The BDPA-CAT has two layers to its architecture, namely the

evaluation system and the user interface.

5.2.1. Evaluation system

Based on the user's inputs to the 21 determinants on the organization, the evaluation system would calculate the BDPA score for the organization by applying Eq. (1). It will also show the organization's BDPA capability level – Low, Medium or High according to the calculated score. BDPA capability level is determined by the range of the score as well as the performance under each determinant group. The determination of capability level will be categorized based on the total score range. For scores below 60, as the maximum average score of each determinant would be 3, or the median of the rating system, the capability level will be categorized as low as it represents that the user's organizations may not have the respective determinant's requirements. Scores above 80 will be categorized as high as the average of each determinant would be 4, representing that the user's organizations have met the requirements of the determinant. or scores between 60 and 80, the capability level will be categorized as medium as the average of each determinant would be in between 3 or 4.

In certain cases, organizations may perform extremely well in one area while neglecting other areas, and yet still able to obtain a “medium” or “high” capability level. To promote well-rounded BDPA development for the organizations, the subtotal score for each determinant group will be checked against the 60% threshold level. The 60% threshold level for the five determinant groups are 11.3%, 14.5%, 7.0%, 9.1%, and 18.0% for DG1, DG2, DG3, DG4 and DG5 respectively. The organization's capability level will directly be evaluated as “Low” if any of the determinant group scores below 60%.

Fig. 3 illustrates the process to determine the capability level through a flow chart diagram.

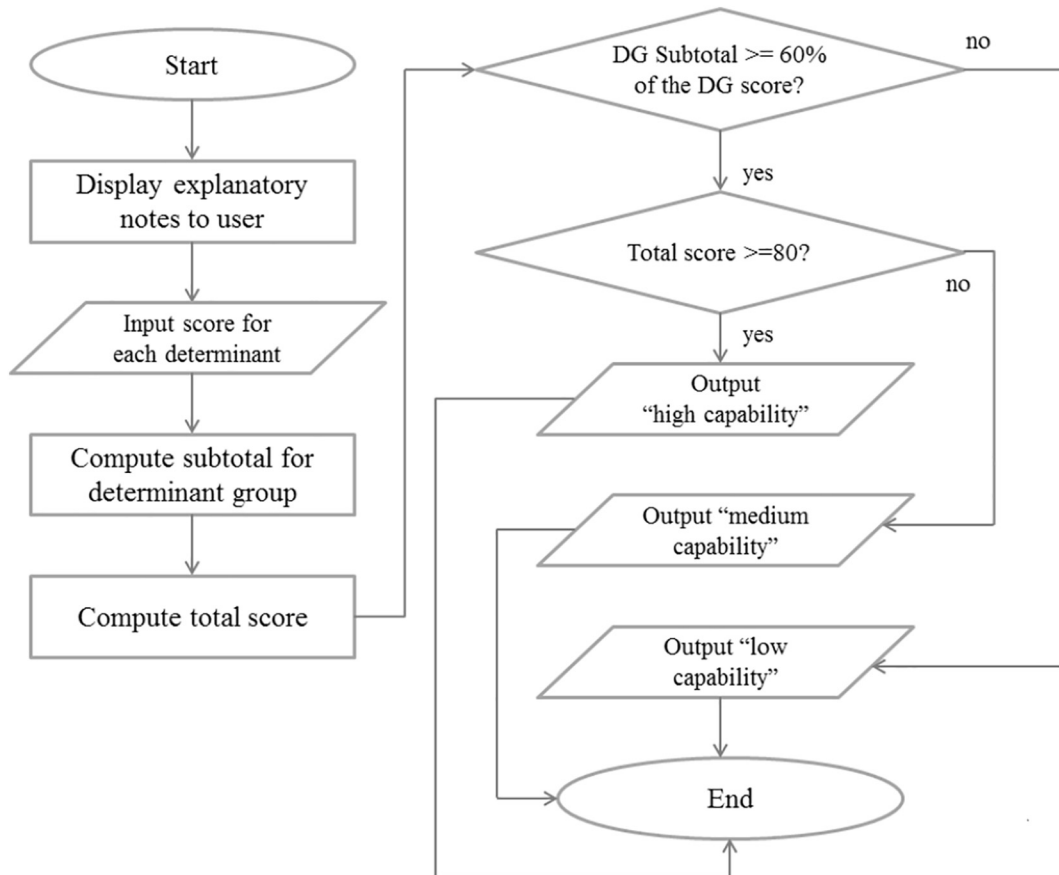


Fig. 3. Flowchart for determining capability level.

	A	B	C	D	E	F
1						
2		Instructions				
3		Purpose: BDPA-CAT aims to assess organization's capability in Big Data and Predictive Analytics implementation and provide feedback regarding organization's BDPA strength and weakness.				
4		Procedures: Please fill in basic information of your organization on this sheet and go to the "Assessment" sheet to rate a total of 21 determinants on a scale of 1-5. You can select the score from the drop down list for each determinant. <i>(1 - strongly disagree, 2 - disagree, 3 - neutral, 4 - agree, 5 - strongly agree)</i>				
5		The BDPA-CAT score will be computed in the Cell D35. BDPA Capability Level will be displayed in a message box when you press ctrl + a on the keyboard.				
6		Profile of Organization				
7		Organization Name		User Name		
8		Business Domain		Position		
9		Organization's years of experience in the construction industry		Organization's years of experience of using predictive data analytics		
10						

Fig. 4. BDPA-CAT instructions for users.

	A	B	C
1			
2		BDPA-CAT Assessment Form	
3	Code	BDPA-CAT Determinants	Rating (1-5)
4	DG1	<i>Data Characteristics</i>	
5	D1	Your organization have high volume of data for analytics. Take 300Tb as the median.	
6	D2	The data for analytics in your organization have high flow rate.	
7	D3	The data for analytics in your organization comes from a variety of sources.	
8	D4	The data for analytics in your organization is reliable.	
9	Subtotal		0.00
10	DG2	<i>BDPA Operations</i>	
11	O1	Your organization has a mature data management and analytics system or a clear SOP for data management and analytics.	
12	O2	Your organization has sufficient funds for investment in BDPA.	
13	O3	Sufficient time can be allocated to BDPA against tight schedules for other projects.	
14	O4	Your organization has a dedicated department / sub-function for data analytics or has plan to form such a department / sub-function in the near future.	
15	O5	Your organization conducts regular review sessions to evaluate data analytics performance.	
16	Subtotal		0.00
17	DG3	<i>Technology and Software</i>	
18	T1	Your organization adopts visualization tools such as a dashboard for users to visualize the analysis processes and results.	
19	T2	Your organization adopts cloud-based platforms for data processing and analytics.	
20	T3	Your organization adopts open source software for BDPA.	
21	T4	Your organization adopts new forms of databases for data storage such as NoSQL.	
22	Subtotal		0.00
23	DG4	<i>Technical Skills and Expertise</i>	
24	S1	Your organization provides technical support for employees while they face difficulties in data analytics software issues.	
25	S2	Your organization conducts relevant data analytics training or provides external training resources for employees.	
26	S3	Employees in your organizations possess high level of competency or skill sets related to predictive analytics.	
27	Subtotal		0.00
28	DG5	<i>Organizational Culture</i>	
29	C1	Your organization has a data-driven culture. Data analysis results are valued and affect major decision making.	
30	C2	Top management in your organization supports innovation and the development of new technologies.	
31	C3	Pursuit of innovation and new technologies aligns with your organization's vision.	
32	C4	Your organization promotes continuous learning and improvement.	
33	C5	Your organization is adaptable to changes.	
34	Subtotal		0
35	Total Score		0.00
36			

Fig. 5. Blank BDPA-CAT assessment form.

5.2.2. User interface

The user interface of the BDPA-CAT consisted of two sections and is displayed in Microsoft Excel. The first section introduces the BDPA-CAT and provides the instructions for using the tool, as shown in Fig. 4. The organization profile can also be included.

In the second section, the users can assess the 21 determinants in the context of their organizations based on a five-point Likert scale as shown in Figs. 5 and 6. A drop down list is used in the input cell to minimize invalid values to be input into the cells.

The computed BDPA-CAT score will be displayed with the respective capability level in a dialogue message box as shown in Fig. 7.

5.3. Validation of BDPA-CAT

Four interviewees representing four construction companies with rich industry experience took part in the validation of BDPA-CAT. The interviewees were requested to assess their organizations according to the BDPA-CAT. The interviewees are independent from the survey respondents and can provide fresh insights and feedbacks regarding the BDPA-CAT. The results are summarized in Table 8.

Company A is a multinational corporation based in United States with presence in 80 countries. Company B is a subsidiary construction firm and specialized in residential projects in Singapore. Company C is

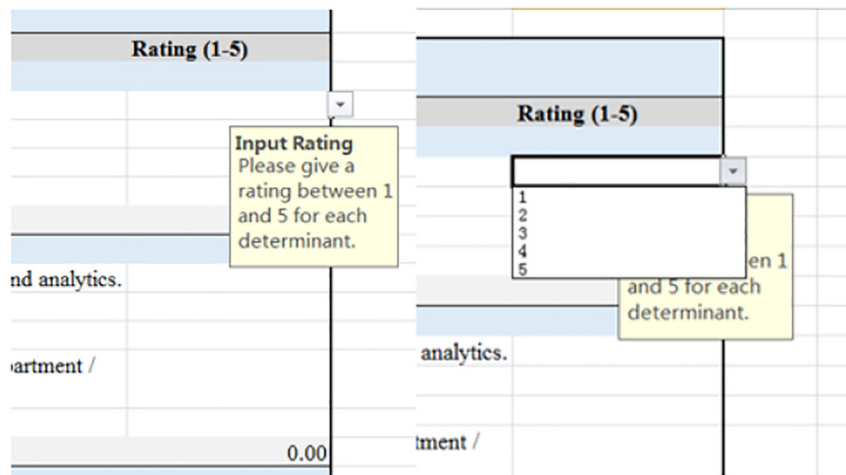


Fig. 6. Pop-up message and drop-down list.

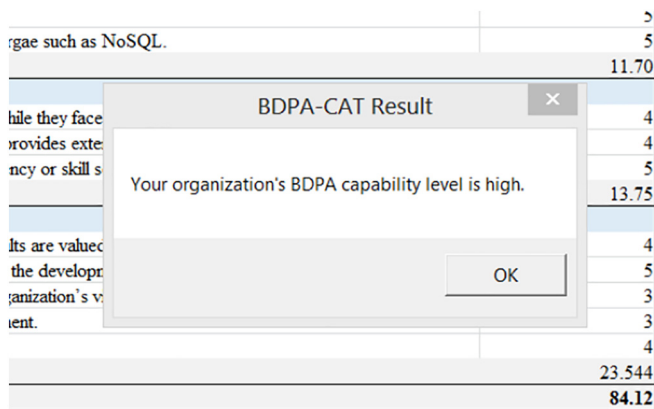


Fig. 7. Sample output result from BDPA-CAT.

a consultancy firm specializing in cost management, project management, project controls and programme management. Company D is a developer specialized in developing condominium projects. Apart from Company B who does not practice predictive analytics, the other companies have experience in predictive analytics at the time of the validation.

Company A has scored more than 80, indicating the high BDPA capability of the organization. This is contributed by the availability of data, presence of mature data management systems, dedicated department for data analytics and having sufficient resources for BDPA. Technological solutions such as Archibus, IBM Tririga and cloud-based platforms were also adopted by the company. However, the technological innovation of the company can be improved. The results indicated that more focus should be placed on the area of organization culture (DG5) to further improve the competency of BDPA.

Company B has a score lower than 60, indicating a low BDPA

capability level. This is contributed by the lack of diversity in data sources, insufficient resources dedicated to BDPA, and the lack of emphasis of BD and innovation by the top management. The results indicated that more focus should be placed on organization operations and technical skills and expertise.

Company C and D scored between 60 and 80, with a medium BDPA capability level. However, Company D has scored lower than the 60% threshold for DG2 and DG3 with a score of 13.16 and 6.29 respectively, and hence was evaluated to have a low BDPA capability level. Company C has no dedicated team for data analytics and provide only basic training for employees. However, the top management expressed strong support for innovation and data analytics and also has a technology support center in place to provide assistance. The results indicate that more attention should be placed on BDPA operations.

Company D has high volume of data from a variety of sources but had no defined SOP or mature data management systems with minimal use of advanced technologies. However, the top management indicated strong support to invest in BDPA with plans to form a dedicated BDPA department and provision of BDPA related trainings. It was understood that Company D had just initiated its plan to implement BDPA. The results indicate that more focus should be placed on BDPA operations and technology and software while the company is implementing BDPA.

The interviewees agreed that the tool has been effective in evaluating the organization's BDPA capability and in providing directions for further improvement. The validation of BDPA-CAT demonstrated that well-rounded performance is required for organizations to have "medium" or "high" BDPA capability levels to ensure that the potential of BDPA implementation can be maximized.

6. Conclusions and recommendations

This study identified the impact of 21 determinants on construction organizations' capability to implement BDPA and the existing

Table 8

Summary of BDPA-CAT output for Company A, B, C and D.

Company	A	B	C	D
Business domain	Consultant	Contractor	Consultant	Developer
BDPA-CAT score	84.1	55.6	72.25	69.8
BDPA capability level	High	Low	Medium	Low*
Strength	Technology and software	Data characteristics	Data characteristics	Organization culture
Area for improvement	Organization culture	Organization operations, technical skills and expertise	Organization operations	Organization operations, technology and software

* The capability level is deemed as "Low" as the organization has scored less than the 60% threshold in at least one determinant group.

capabilities of construction organizations to adopt BDPA. It was found that construction organizations do have the capability to implement BDPA according to the determinants and that all five determinant groups are important. Their order of importance was determined by the mean rank analysis - in a descending order - DG5, DG2, DG1, DG4 and DG3. Using the survey results as a reference, a BDPA capability scoring approach was developed to assess a construction organizations' BDPA capability. The BDPA-CAT was developed using Microsoft Excel and VBA, based on the BDPA capability scoring approach, which was validated with four construction organizations.

This research is limited in a few areas. Firstly, the sample size of the questionnaire is relatively small, although there is wide coverage in terms of business domains and respondents' positions. A larger sample size may produce more reliable results in the determining of impact of BDPA capability determinants. Next, as the survey is conducted through an online platform, respondents are unable to clarify doubts and misinterpretation of concepts immediately. To mitigate this problem, explanatory notes for certain survey questions were provided to show definitions or examples. The online survey platform also supports pictures and diagrams to better illustrate complicated concepts and techniques. It could actively reduce the chance of misinterpretation. In addition, the survey questionnaire was designed to conduct a general assessment of the organizations' existing capability and readiness to implement BDPA and did not assess the specific sources of big data, usability of the data available in the organizations and the specific uses of tools provided in the technologies and software. These can be investigated in future studies.

This study provided 21 determinants that impact construction organizations' capability to implement BDPA. It is the first study that presents these determinants for the construction industry and adds to the existing body of knowledge. On top of that, the BDPA-CAT is also the first computerized assessment tool developed to assess BDPA capabilities of construction organizations. The developed BDPA-CAT also contributes to practice in that construction organizations can evaluate their BDPA capabilities and identify areas of improvement in order to maximize the potential of BDPA. The BDPA-CAT can be used in other geographical locations beyond Singapore, as the determinants identified are more related to the organizational processes and technical skills, which are also applicable in other countries and regions. However, the weightages for each determinant should be reviewed using the approach proposed in this study, as the impact of each determinant may be perceived differently from Singapore.

As the practice of BDPA in Singapore's construction industry is still in its infancy, new determinants that impact organizations' BDPA capability may emerge as more firms adopt BDPA. Further reviews and updates of the BDPA-CAT can be conducted. At the same time, best practices in implementing BDPA can also be further studied to improve the BDPA-CAT in future studies. This study specifically focuses on the aspect of predictive analytics. As technology advances and volume, variety and velocity of data grow, prescriptive analytics which explores optimal solutions and creates more value to organizations can be further researched. In addition, specific sources of BD, usability of the data available in organizations and specific uses of the tools available in technologies and software adopted by the organizations and their impact on BDPA capabilities can be investigated in future studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Wikibon, SiliconANGLE, Global Big Data Market Size 2011–2027, Statista, 2018, <https://www.statista.com/statistics/254266/global-big-data-market-forecast/>,

- Accessed date: 2 May 2019.
- [2] F. Provost, T. Fawcett, *Data Science for Business: What you Need to Know about Data Mining and Data-Analytic Thinking*, O'Reilly, 2013 (ISBN:9781449361327).
- [3] F.T. Matsunaga, J.D. Brancher, R.M. Busto, Data Mining Applications and Techniques: A Systematic Review, (2015), <https://doi.org/10.5281/ZENODO.59454>.
- [4] K. Ballard-Bloomfield, D. Brook, How can big data enable more efficient and effective location decisions and planning? - area development, (2018) <https://www.areadevelopment.com/corporate-site-selection-factors/Q3-2018/big-data-efficient-effective-location-decisions-and-planning.shtml> (accessed May 30, 2019).
- [5] V.T. Luu, S.-Y. Kim, N. Van Tuan, S.O. Ogunlana, Quantifying schedule risk in construction projects using Bayesian belief networks, *Int. J. Proj. Manag.* 27 (2009) 39–50, <https://doi.org/10.1016/j.ijproman.2008.03.003>.
- [6] A. González-Vidal, V. Moreno-Cano, F. Terroso-Sáenz, A.F. Skarmeta, Towards Energy Efficiency Smart Buildings Models Based on Intelligent Data Analytics, *Procedia Computer Science*, Elsevier, 2016, pp. 994–999, <https://doi.org/10.1016/j.procs.2016.04.213>.
- [7] M. Bilal, L.O. Oyedele, J. Qadir, K. Munir, S.O. Ajayi, O.O. Akinade, H.A. Owolabi, H.A. Alaka, M. Pasha, Big Data in the construction industry: a review of present status, opportunities, and future trends, *Adv. Eng. Inform.* 30 (2016) 500–521, <https://doi.org/10.1016/j.aei.2016.07.001>.
- [8] S.M.E. Sepasgozar, M. Loosemore, S.R. Davis, Conceptualising information and equipment technology adoption in construction: a critical review of existing research, *Architectural Management* 23 (2016) 158–176, <https://doi.org/10.1108/ECAM-05-2015-0083>.
- [9] McKinsey Global Institute, Big Data: The Next Frontier for Innovation, Competition and Productivity, www.mckinsey.com/mgi, (2011), Accessed date: 24 April 2019.
- [10] M.A. Beyer, D. Laney, The importance of “big data”: A definition, (2012) <https://www.gartner.com/en/documents/2057415/the-importance-of-big-data-a-definition> (accessed April 24, 2019).
- [11] C. Chen, Storey, H. Chen, R.H.L. Chiang, V.C. Storey, Business intelligence and analytics: from big data to big impact, *MIS Q.* 36 (2012) 1165, <https://doi.org/10.2307/41703503>.
- [12] P. Veras, S. Suresh, S. Renukappa, The adoption of big data concepts for sustainable practices implementation in the construction industry, 2018 IEEE/ACM International Conference on Utility and Cloud Computing Companion (UCC Companion), IEEE, 2019, pp. 349–352, <https://doi.org/10.1109/UCC-Companion.2018.00079>.
- [13] Oracle, What is Big Data? <https://www.oracle.com/big-data/guide/what-is-big-data.html>, (2019), Accessed date: 8 May 2019.
- [14] IBM, What Is Big Data? - Bringing Big Data to the Enterprise, <http://www-01.ibm.com/software/data/bigdata/>, (2013), Accessed date: 30 April 2019.
- [15] R.M. Losee, Browsing mixed structured and unstructured data, *Inf. Process. Manag.* 42 (2006) 440–452, <https://doi.org/10.1016/j.ipm.2005.02.001>.
- [16] H. Baars, H.-G. Kemper, Management support with structured and unstructured data—an integrated business intelligence framework, *Inf. Syst. Manag.* 25 (2008) 132–148, <https://doi.org/10.1080/10580530801941058>.
- [17] J.R. Evans, C.H. Lindner, Business analytics: the next frontier for decision sciences, decision line, http://faculty.cbpp.uaa.alaska.edu/afef/business_analytics.htm, (2012).
- [18] G. Wang, A. Gunasekaran, E.W.T. Ngai, T. Papadopoulos, Big data analytics in logistics and supply chain management: certain investigations for research and applications, *Int. J. Prod. Econ.* 176 (2016) 98–110, <https://doi.org/10.1016/j.ijpe.2016.03.014>.
- [19] A. Banerjee, T. Bandyopadhyay, P. Acharya, Data analytics: hyped up aspirations or true potential? Vikalpa: The Journal for Decision Makers 38 (2013) 1–12, <https://doi.org/10.1177/0256090920130401>.
- [20] J. Fan, F. Han, H. Liu, Challenges of big data analysis, *Natl. Sci. Rev.* 1 (2014) 293–314, <https://doi.org/10.1093/nsr/nwt032>.
- [21] T. Hastie, J. Friedman, R. Tibshirani, *The Elements of Statistical Learning*, Springer New York, New York, NY, 2001, <https://doi.org/10.1007/978-0-387-21606-5>.
- [22] D. Dietrich, R. Heller, B. Yang, *EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data*, (2015) (ISBN:9781118876138).
- [23] SAS Insights, Predictive analytics: what it is and why it matters (n.d.), https://www.sas.com/en_sg/insights/analytics/predictive-analytics.html, Accessed date: 29 April 2019.
- [24] A. Oudjehane, S. Moeini, Big Data in Construction Projects: Risk and Opportunity Management, http://pmsymposium.umd.edu/pm2017/wp-content/uploads/sites/3/2017/01/Big_Data_in_Construction_Projects_Risk_and_Opportunity_Management-Moeini.pdf, (2017), Accessed date: 11 June 2019.
- [25] Y. Zhang, H. Luo, Y. He, A System for Tender Price Evaluation of Construction Project Based on Big Data, *Procedia Engineering*, Elsevier, 2015, pp. 606–614, <https://doi.org/10.1016/j.proeng.2015.10.114>.
- [26] H. Alaka, L. Oyedele, H. Owolabi, O. Akinade, M. Bilal, S. Ajayi, A big data analytics approach for construction firms failure prediction models, *IEEE Trans. Eng. Manag.* (2018) 1–10, <https://doi.org/10.1109/TEM.2018.2856376>.
- [27] S.M. Trost, G.D. Oberlender, Predicting accuracy of early cost estimates using factor analysis and multivariate regression, *J. Constr. Eng. Manag.* 129 (2003) 198–204, [https://doi.org/10.1061/\(ASCE\)0733-9364\(2003\)129:2\(198\)](https://doi.org/10.1061/(ASCE)0733-9364(2003)129:2(198)).
- [28] T.P. Williams, J. Gong, Predicting construction cost overruns using text mining, numerical data and ensemble classifiers, *Autom. Constr.* 43 (2014) 23–29, <https://doi.org/10.1016/j.autcon.2014.02.014>.
- [29] S.U. Han, S.H. Lee, F. Peña-Mora, A Machine-Learning Classification Approach to Automatic Detection of Workers' Actions for Behavior-Based Safety Analysis, *Computing in Civil Engineering 2012 American Society of Civil Engineers*, Reston,

- VA, 2012, pp. 65–72, <https://doi.org/10.1061/9780784412343.0009>.
- [30] W. Lu, X. Chen, Y. Peng, L. Shen, Benchmarking construction waste management performance using big data, *Resources, Conservation & Recycling* 105 (2015) 49–58, <https://doi.org/10.1016/j.resconrec.2015.10.013>.
- [31] H.M. Chen, K.C. Chang, T.H. Lin, A cloud-based system framework for performing online viewing, storage, and analysis on big data of massive BIMs, *Autom. Constr.* 71 (2016) 34–48, <https://doi.org/10.1016/j.autcon.2016.03.002>.
- [32] K.K. Han, M. Golparvar-Fard, Potential of big visual data and building information modeling for construction performance analytics: an exploratory study, *Autom. Constr.* 73 (2017) 184–198, <https://doi.org/10.1016/j.autcon.2016.11.004>.
- [33] A. Tibaut, D. Zazula, Sustainable management of construction site big visual data, *Sustain. Sci.* 13 (2018) 1311–1322, <https://doi.org/10.1007/s11625-018-0595-9>.
- [34] C. Miller, F. Meggers, Mining electrical meter data to predict principal building use, performance class, and operations strategy for hundreds of non-residential buildings, *Energy and Buildings* 156 (2017) 360–373, <https://doi.org/10.1016/j.enbuild.2017.09.056>.
- [35] D. Lindelöf, M. Alisafae, P. Borsò, C. Grigis, X. Mocellin, J. Viaene, Bayesian evaluation of energy conservation measures: a case study with a model-predictive controller for space heating on a commercial building, *Energy Procedia*, Elsevier, 2017, pp. 235–240, <https://doi.org/10.1016/j.egypro.2017.07.351>.
- [36] K. Song, N. Kwon, K. Anderson, M. Park, H.S. Lee, S.H. Lee, Predicting hourly energy consumption in buildings using occupancy-related characteristics of end-user groups, *Energy and Buildings* 156 (2017) 121–133, <https://doi.org/10.1016/j.enbuild.2017.09.060>.
- [37] L. Linder, D. Vionnet, J.P. Bacher, J. Hennebert, Big building data—a big data platform for smart buildings, *Energy Procedia*, Elsevier, 2017, pp. 589–594, <https://doi.org/10.1016/j.egypro.2017.07.354>.
- [38] B.T. Hazen, C.A. Boone, J.D. Ezell, L.A. Jones-Farmer, Data quality for data science, predictive analytics, and big data in supply chain management: an introduction to the problem and suggestions for research and applications, *Int. J. Prod. Econ.* 154 (2014) 72–80, <https://doi.org/10.1016/j.ijpe.2014.04.018>.
- [39] B.T. Hazen, R.E. Overstreet, C.G. Cegielski, Supply chain innovation diffusion: going beyond adoption, *The International Journal of Logistics Management* 23 (2012) 119–134, <https://doi.org/10.1108/09574091211226957>.
- [40] M. Gupta, J.F. George, Toward the development of a big data analytics capability, *Inf. Manag.* 53 (2016) 1049–1064, <https://doi.org/10.1016/j.im.2016.07.004>.
- [41] B.H. Wixom, H.J. Watson, An empirical investigation of the factors affecting data warehousing success, *MIS Q.* 25 (2001) 17, <https://doi.org/10.2307/3250957>.
- [42] F.J. Mata, W.L. Fuerst, J.B. Barney, Information technology and sustained competitive advantage: a resource-based analysis, *MIS Q.* 19 (1995) 487, <https://doi.org/10.2307/249630>.
- [43] T. H. Davenport, How strategists use “big data” to support internal business decisions, discovery and production, *Strategy and Leadership* 42 (2014) 45–50, <https://doi.org/10.1108/SL-05-2014-0034>.
- [44] A. Carmeli, A. Tishler, Resources, capabilities, and the performance of industrial firms: a multivariate analysis, *Manag. Decis. Econ.* 25 (2004) 299–315, <https://doi.org/10.1002/mde.1192>.
- [45] T.H. Davenport, D. Patil, Data scientist: the sexiest job of the 21st century, *Harvard business review*, <http://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century/ar/pr>, (2012).
- [46] T.H. Davenport, J. Dyché, Big Data in Big Companies, https://docs.media.bitpipe.com/io_10x/io_102267/item_725049/Big-Data-in-Big-Companies.pdf, (2013), Accessed date: 30 April 2019.
- [47] H. Liang, N. Saraf, Q. Hu, Y. Xue, Assimilation of enterprise systems: the effect of institutional pressures and the mediating role of top management, *MIS Q.* 31 (2007) 59–87, <https://doi.org/10.2307/25148781>.
- [48] G.D. Bhatt, V. Grover, Types of information technology capabilities and their role in competitive advantage: an empirical study, *J. Manag. Inf. Syst.* 22 (2005) 253–277, <https://doi.org/10.1080/07421222.2005.11045844>.
- [49] Z. Bi, D. Cochran, Big data analytics with applications, *Journal of Management Analytics* 1 (2014) 249–265, <https://doi.org/10.1080/23270012.2014.992985>.
- [50] A. Darko, A.P.C. Chan, E.E. Ameyaw, B.-J. He, A.O. Olanipekun, Examining issues influencing green building technologies adoption: the United States green building experts’ perspectives, *Energy and Buildings* 144 (2017) 320–332, <https://doi.org/10.1016/j.enbuild.2017.03.060>.
- [51] Y.Y. Li, P.-H. Chen, D.A.S. Chew, C.C. Teo, R.G. Ding, Critical Project Management factors of AEC firms for delivering green building projects in Singapore, *J. Constr. Eng. Manag.* 137 (2011) 1153–1163, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000370](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000370).
- [52] P.L. Alreck, R.B. Settle, *The Survey Research Handbook*, McGraw-Hill/Irwin, 9780072945485, 2004.
- [53] R.L. Ott, M. Longnecker, *An Introduction to Statistical Methods and Data Analysis*, Seventh, Cengage Learning, 2016 (ISBN:9781305269477).
- [54] X. Zhao, B.G. Hwang, Y. Gao, A fuzzy synthetic evaluation approach for risk assessment: a case of Singapore’s green projects, *J. Clean. Prod.* 115 (2016) 203–213, <https://doi.org/10.1016/j.jclepro.2015.11.042>.
- [55] B.G. Hwang, M. Shan, K.Y. Looi, Key constraints and mitigation strategies for prefabricated prefinished volumetric construction, *J. Clean. Prod.* 183 (2018) 183–193, <https://doi.org/10.1016/j.jclepro.2018.02.136>.
- [56] X. Zhao, B.G. Hwang, S.P. Low, Critical success factors for enterprise risk management in Chinese construction companies, *Constr. Manag. Econ.* 31 (2013) 1199–1214, <https://doi.org/10.1080/01446193.2013.867521>.
- [57] A.J.-P. Tixier, M.R. Hallowell, A. Albert, L. van Boven, B.M. Kleiner, Psychological antecedents of risk-taking behavior in construction, *J. Constr. Eng. Manag.* 140 (2014) 04014052, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000894](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000894).
- [58] X. Zhao, B.-G. Hwang, S.P. Low, An enterprise risk management knowledge-based decision support system for construction firms, *Eng. Constr. Archit. Manag.* 23 (2016) 369–384, <https://doi.org/10.1108/ECAM-03-2015-0042>.
- [59] B.-G. Hwang, M. Shan, K.-Y. Looi, Knowledge-based decision support system for prefabricated prefinished volumetric construction, *Autom. Constr.* 94 (2018) 168–178, <https://doi.org/10.1016/j.autcon.2018.06.016>.
- [60] M. Loosemore, Improving construction productivity: a subcontractor’s perspective, *Eng. Constr. Archit. Manag.* 21 (2014) 245–260, <https://doi.org/10.1108/ECAM-05-2013-0043>.
- [61] R.V. Vargas, Using the analytic hierarchy process (AHP) to select and prioritize projects in a portfolio, *PMI Global Congress*, 2010, pp. 1–22 <https://www.pmi.org/learning/library/analytic-hierarchy-process-prioritize-projects-6608>, Accessed date: 9 August 2019.